Project: Worked examples
Title: Cylinder and actuator circuit
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1. Scope and method

A double-acting hydraulic cylinder in its circuit: the bore, rod and annulus areas; the force and speed on the extend and the retract stroke with mechanical and volumetric efficiency and an optional back pressure, or with the regenerative extend connection; the swept volume and the cycle time; the pump delivery, shaft power, torque and speed; a hydraulic motor's torque; the velocity in the suction, pressure and return lines against your own limits and the smallest listed hose or pipe bore that meets them; a sharp orifice's flow for flow-control screening; and a screening check of the rod against buckling (Euler above the transition slenderness, Johnson parabola below it). screening only; ISO/TS 13725 method not implemented. Not a cylinder design to a standard: the limits, the hose sizes, the mounting factor and the material properties are your records.

WHAT IT CALCULATES	RELATION
Piston and rod areas	$A_b = \pi \cdot D^2/4$, $A_r = \pi \cdot d_r^2/4$, $A_a = A_b - A_r$, area ratio $\varphi = A_b/A_a$
Cylinder force	$F_{ext} = (p \cdot A_b - p_{back} \cdot A_a) \cdot \eta_m$, $F_{ret} = (p \cdot A_a - p_{back} \cdot A_b) \cdot \eta_m$; regenerative extend $F_{ext} = p \cdot A_r \cdot \eta_m$
Speed and time	$v_{ext} = Q \cdot \eta_{vc}/A_b$ (regenerative: A_r), $v_{ret} = Q \cdot \eta_{vc}/A_a$; t = stroke/v; $t_{cycle} = t_{ext} + t_{ret} + t_{dwell}$
Required speed (optional)	when v_{req} is given it gives the flow, $Q = v_{req} \cdot A_b/\eta_{vc}$ (regenerative: A_r), and replaces the entered Q; the flow must lie within 0.1 to 2000 L/min
Swept volume	$\Delta V_{ext} = A_b \cdot$stroke (regenerative: $A_r \cdot$stroke), $\Delta V_{ret} = A_a \cdot$stroke; ΔV = the larger
Pump flow and power	$Q_p = V_g \cdot n \cdot \eta_{vp}$; $P = p \cdot Q_p/\eta_{tp}$; $T = P/\omega$; displacement needed $V_{g,req} = Q/(n \cdot \eta_{vp})$
Hydraulic motor	$T = \Delta p \cdot V_g/(2\pi) \cdot \eta_{hm}$ (V_g per revolution)
Line velocities	$v = Q_{line}/A_{line}$; suction $Q_{suc} = \max(Q, V_g \cdot n)$ (the geometric delivery), pressure/cap-port line $Q_{prs} = Q_{line} \cdot A_b/A_a$ on the retract stroke (regenerative extend: $Q_{line} \cdot A_b/A_r$ if larger), return $Q_{rtn} = Q_{line} \cdot A_b/A_a$, with $Q_{line} = \max(Q, Q_p)$; smallest bore $d = \sqrt{(4Q/(\pi \cdot v_{max}))}$, recommended = the smallest listed size not below it
Orifice flow	$Q = C_d \cdot A_o \cdot \sqrt{(2\Delta p/\rho)}$; bore for the circuit flow $d_o = \sqrt{(4Q/(\pi \cdot C_d \cdot \sqrt{(2\Delta p/\rho)}))}$
Rod buckling (screening)	buckling length L_b = the pin-to-pin length at full stroke = L_{closed} + stroke (L_{closed} blank: estimated as 2·stroke + L_{add}, the body taken as long as the stroke), $L_{eff} = K \cdot L_b$, $r = d_r/4$, $\lambda = L_{eff}/r$, $\lambda_c = \pi \cdot \sqrt{(2E/f_y)}$; Euler $F_{cr} = \pi^2 \cdot E \cdot I/L_{eff}^2$ for $\lambda > \lambda_c$, Johnson $F_{cr} = A_r \cdot f_y \cdot [1 - f_y \cdot \lambda^2/(4\pi^2 E)]$ for $\lambda \le \lambda_c$; load $F_b = \max(F_{push}, F_{stall})$ with $F_{stall} = p \cdot A_b$ (the stalled cylinder at th...

Basis: Cylinder buckling load: method not implemented, screening only[1]; Force, speed, flow and power relations, closed form[2](Cylinder and pump chapters).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
mode	Extend connection	Standard		assumption: Standard connection: the ram needs its full bore force for the press stroke.	any
p	Supply pressure at the cylinder and the pump (gauge)	180	bar g	spec: Hydraulic specification HS-31: relief valve set 180 bar at the ram	1 … 700 bar g
p_{back}	Back pressure on the discharge side (standard connection, gauge)	0	bar g	assumption: Return line direct to tank; the return filter drop is neglected for this example.	0 … 700 bar g
Q	Flow into the cylinder (blank when a required extend speed gives it)	60	L/min	spec: Hydraulic specification HS-31: ram flow 60 L/min	0.1 … 2000 L/min
ρ	Fluid density	870	kg/m³	reference: fluid data sheet: mineral oil, density at 40 °C, to be taken from the oil supplier's data	700 … 1300 kg/m³
F_{push}	Load to push on the extend stroke (compresses the rod)	150	kN	spec: Hydraulic specification HS-31: pressing force 150 kN	0 … 15000 kN
F_{pull}	Load to pull on the retract stroke	60	kN	spec: Hydraulic specification HS-31: return stroke load 60 kN	0 … 15000 kN
D	Bore	125	mm	drawing: DWG-4310 rev B	10 … 500 mm
d_r	Rod diameter	90	mm	drawing: DWG-4310 rev B	4 … 450 mm
stroke	Stroke	800	mm	drawing: DWG-4310 rev B	1 … 5000 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
η_m	Mechanical efficiency of the cylinder (seal and guide friction)	0.95		assumption: Typical seal and guide friction for a new cylinder; to be confirmed against the cylinder maker's figure.	0.5 … 1
η_{vc}	Volumetric efficiency of the cylinder (leakage past the piston)	0.98		assumption: Allowance for leakage past the piston; to be confirmed against the cylinder maker's figure.	0.5 … 1
t_{dwell}	Dwell time added to each cycle	0	s	assumption: Cycle time of the strokes only; dwell added by the machine controller.	0 … 3600 s
K_m	Effective-length factor K of the mounting (L_{eff} = K·L)	1		reference: record: cylinder mounting (clevis at both ends, pinned), to be taken from the cylinder maker's mounting guidance	0.5 … 2.5
L_{closed}	Closed (retracted) pin-to-pin length of the cylinder, from the cap-end mounting to the rod-end mounting (the cylinder body is part of the buckling length); blank = estimated as stroke + L_{add}	950	mm	drawing: DWG-4310 rev B	1 … 20000 mm
L_{add}	Length of the closed cylinder beyond its stroke (body, end caps and mounting eyes), used only when L_{closed} is blank: the closed length is then stroke + L_{add}	150	mm	drawing: DWG-4310 rev B	0 … 5000 mm
E	Young's modulus of the rod	210	GPa	reference: record: rod material (chromed carbon steel), to be taken from the material library	50 … 250 GPa

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
f_y	Yield strength of the rod	500	MPa	reference: record: rod material (chromed carbon steel), to be taken from the rod maker's certificate	100 … 1500 MPa
S_f	Safety factor on the critical load	3.5		assumption: Safety factor chosen for this example; the design reference sets the real one.	1 … 20
V_g	Pump displacement per revolution (leave blank to skip the pump delivery check)	45	cm³/rev	spec: Hydraulic specification HS-31: fixed-displacement pump 45 cm³/rev	0.1 … 1000 cm³/rev
n_p	Pump speed	1450	rpm	spec: Hydraulic specification HS-31: 4-pole motor, 1450 rpm	50 … 6000 rpm
η_{vp}	Volumetric efficiency of the pump	0.95		assumption: Pump volumetric efficiency at 180 bar; to be taken from the pump data sheet.	0.5 … 1
η_{tp}	Total efficiency of the pump (volumetric × hydromechanical)	0.88		assumption: Pump total efficiency at 180 bar; to be taken from the pump data sheet.	0.5 … 1
η_{hm}	Hydromechanical efficiency of the motor	0.9		default	0.5 … 1
d_{suc}	Suction line inside diameter	50	mm	drawing: DWG-4311 rev A	3 … 200 mm
d_{prs}	Pressure line inside diameter (to the cylinder cap port; judged at the larger flow of the two strokes)	25	mm	drawing: DWG-4311 rev A	3 … 200 mm
d_{ret}	Return line inside diameter	32	mm	drawing: DWG-4311 rev A	3 … 200 mm
$v_{suc,max}$	Suction line velocity limit (your record; the default is a placeholder, not a code value)	1.2	m/s	reference: record: line velocity limits, a placeholder to be replaced from the design reference	0.1 … 20 m/s

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
$v_{prs,max}$	Pressure line velocity limit (your record; the default is a placeholder, not a code value)	5	m/s	reference: record: line velocity limits, a placeholder to be replaced from the design reference	0.1 ... 20 m/s
$v_{ret,max}$	Return line velocity limit (your record; the default is a placeholder, not a code value)	3	m/s	reference: record: line velocity limits, a placeholder to be replaced from the design reference	0.1 ... 20 m/s
hoses	Available hose or pipe inside diameters (from your supplier's range)	7 row(s)		reference: record: illustrative hose size list invented for this example, to be replaced by the supplier's range	any
C_d	Orifice discharge coefficient	0.62		default	0.2 ... 1

TABLE ROW	VALUES
hoses[1]	ID = 16 mm
hoses[2]	ID = 19 mm
hoses[3]	ID = 25 mm
hoses[4]	ID = 32 mm
hoses[5]	ID = 38 mm
hoses[6]	ID = 51 mm
hoses[7]	ID = 63 mm

3. Assumptions

A1. The supply pressure p is the gauge pressure at the cylinder port and, for the pump, at the pump outlet; line losses are not subtracted (the line losses of a pipe run come from the pipe calculation). The force is the pressure on the area less the back pressure on the other side, times the mechanical efficiency. ($F = p \cdot A \cdot \eta_m$, closed form.)
A2. The speed is the flow into the cylinder times the volumetric efficiency over the area, in steady motion: no acceleration, cushioning or compressibility. The cycle time is the two strokes plus the dwell. ($v = Q \cdot \eta_{vc} / A$.)
A3. The pump power is the supply pressure times the pump's delivery over the total efficiency: when the pump delivers more than the circuit takes, the excess is taken to go over the relief valve at the supply pressure, which is the conservative case for the drive. The shaft torque is the power over the angular speed. ($P = p \cdot Q_p / \eta_{tp}$.)
A4. The line flows use the pump delivery when it is larger than the circuit flow: the suction line the geometric delivery $V_g \cdot n$, the pressure line and the return line $Q_{line} = \max(Q, Q_p)$, which carries no volumetric efficiency while the speeds use $Q \cdot \eta_{vc}$ (the flows are the conservative side of the speeds). The line to the cylinder cap port is judged at the larger flow of the two strokes: on the retract stroke it carries the cap-side discharge $Q_{line} \cdot A_b / A_a$ (the same flow as the return line), and in the regenerative extend $Q_{line} \cdot A_b / A_r$ if that is larger. The velocity limits are your records; the sheet's defaults are placeholders, not values from a standard. (Continuity.)

A5. The orifice relation is the sharp-edged turbulent one, $Q = C_d \cdot A \cdot \sqrt{(2\Delta p/\rho)}$, with your discharge coefficient; it ignores viscosity and cavitation and is a screening of a flow-control orifice, not a valve selection. (Orifice equation, closed form.)
A6. The rod is treated as a solid round column of its own diameter with the effective-length factor of the mounting; the tube, the piston and the rod-end stiffness are not modelled, and the rod's diameter is taken over the whole pin-to-pin length at full stroke (closed length plus stroke, the body included; a stiffer body than the rod makes this conservative). The rod is judged at the larger of the push load you enter and the stall thrust $p \cdot A_b$ (a stalled, obstructed or relief-limited cylinder loads the rod with the full pressure force on the bore: no flow, so no back pressure, and the seal friction does not relieve it); no pressure-limiting device is credited, and the regenerative extend is judged at the same stall thrust because the standard connection can always be selected. The mounting factor K_m defaults to 1 (pinned at both ends): a fixed-free mounting is $K_m = 2$, and the factor and the safety factor S_f are your records. screening only; ISO/TS 13725 method not implemented. (Euler and Johnson column relations, closed form.)
A7. mode (Extend connection) is taken as Standard. (Standard connection: the ram needs its full bore force for the press stroke.)
A8. p_{back} (Back pressure on the discharge side (standard connection, gauge)) is taken as 0 bar g. (Return line direct to tank; the return filter drop is neglected for this example.)
A9. η_m (Mechanical efficiency of the cylinder (seal and guide friction)) is taken as 0.95. (Typical seal and guide friction for a new cylinder; to be confirmed against the cylinder maker's figure.)
A10. η_{vc} (Volumetric efficiency of the cylinder (leakage past the piston)) is taken as 0.98. (Allowance for leakage past the piston; to be confirmed against the cylinder maker's figure.)
A11. t_{dwell} (Dwell time added to each cycle) is taken as 0 s. (Cycle time of the strokes only; dwell added by the machine controller.)
A12. S_f (Safety factor on the critical load) is taken as 3.5. (Safety factor chosen for this example; the design reference sets the real one.)
A13. η_{vp} (Volumetric efficiency of the pump) is taken as 0.95. (Pump volumetric efficiency at 180 bar; to be taken from the pump data sheet.)
A14. η_{tp} (Total efficiency of the pump (volumetric × hydromechanical)) is taken as 0.88. (Pump total efficiency at 180 bar; to be taken from the pump data sheet.)
A15. Standard connection: the extend force is the pressure on the bore area less the back pressure on the annulus area; the retract force the pressure on the annulus less the back pressure on the bore. (Continuity and force balance.)

4. Working

[28] $A_b = \pi * D^2 / 4$
$= \pi \times (125\ mm)^2 / 4$
$= 122.72\ cm^2$
Bore (cap-side) area
[29] $A_r = \pi * d_r^2 / 4$
$= \pi \times (90\ mm)^2 / 4$
$= 63.617\ cm^2$
Rod area
[30] $A_a = A_b - A_r$
$= 122.72\ cm^2 - 63.617\ cm^2$
$= 59.101\ cm^2$
Annulus (rod-side) area
[31] $\varphi = A_b / A_a$
$= 122.72\ cm^2 / 59.101\ cm^2$
$= 2.0764$
Area ratio A_b/A_a
[33] $F_{ext} = (p * A_b - p_{back} * A_a) * \eta_m$
$= (180\ bar\ g \times 122.72\ cm^2 - 0\ bar\ g \times 59.101\ cm^2) \times 0.95$
$= 209.85\ kN$

Extend force

[34] $v_{ext} = Q * \eta_{vc} / A_b$

$= 60\ L/min \times 0.98 / 122.72\ cm^2$

$= 79.858\ mm/s$

Extend speed

[35] $dV_{ext} = A_b * stroke$

$= 122.72\ cm^2 \times 800\ mm$

$= 9.8175\ L$

Liquid taken from the supply for the extend stroke

[36] $F_{ret} = (p * A_a - p_{back} * A_b) * \eta_m$

$= (180\ bar\ g \times 59.101\ cm^2 - 0\ bar\ g \times 122.72\ cm^2) \times 0.95$

$= 101.06\ kN$

Retract force

[37] $v_{ret} = Q * \eta_{vc} / A_a$

$= 60\ L/min \times 0.98 / 59.101\ cm^2$

$= 165.82\ mm/s$

Retract speed

[38] $t_{ext} = stroke / v_{ext}$

$= 800\ mm / 79.858\ mm/s$

$= 10.018\ s$

Extend stroke time

[39] $t_{ret} = stroke / v_{ret}$

$= 800\ mm / 165.82\ mm/s$

$= 4.8246\ s$

Retract stroke time

[40] $t_{cycle} = t_{ext} + t_{ret} + t_{dwell}$

$= 10.018\ s + 4.8246\ s + 0\ s$

$= 14.842\ s$

Cycle time

[41] $dV_{ret} = A_a * stroke$

$= 59.101\ cm^2 \times 800\ mm$

$= 4.7281\ L$

Liquid taken from the supply for the retract stroke

[42] $dV = \max(dV_{ext}, dV_{ret})$

$= \max(9.8175\ L, 4.7281\ L)$

$= 9.8175\ L$

Swept volume of the larger stroke

[43] $Q_p = V_g * n_p * \eta_{vp}$

$= 45\ cm^3/rev \times 1450\ rpm \times 0.95$

$= 61.988\ L/min$

Pump delivery

[44] $Vg_{req} = Q / (n_p * \eta_{vp})$

$= 60\ L/min / (1450\ rpm \times 0.95)$

$= 43.557\ cm^3/rev$

Pump displacement the circuit flow needs at this speed

[45] $u_{pump} = Q / Q_p$

$= 60\ L/min / 61.988\ L/min$

$= 0.96794$

Circuit flow over the pump delivery

[46] $Q_{pump} = Q_p$

$= 61.988\ L/min$

$= 61.988\ L/min$

Flow the pump delivers at the supply pressure

[47] $P_{hyd} = p * Q_{pump}$
= 180 bar g × 61.988 L/min
= 18.596 kW
Hydraulic power at the pump outlet
[48] $P_{pump} = P_{hyd} / \eta_{tp}$
= 18.596 kW / 0.88
= 21.132 kW
Pump shaft power
[49] $T_{shaft} = P_{pump} / n_p$
= 21.132 kW / 1450 rpm
= 139.17 N·m
Pump shaft torque P / ω
[50] $n_{shaft} = n_p$
= 1450 rpm
= 1450 rpm
Pump shaft speed
[51] $Q_{line} = max(Q, Q_p)$
= max(60 L/min, 61.988 L/min)
= 61.988 L/min
Flow in the pressure and return lines before the area ratio (the larger of the circuit flow and the pump delivery)
[52] $Q_{suc} = max(Q, V_g * n_p)$
= max(60 L/min, 45 cm³/rev × 1450 rpm)
= 65.25 L/min
Flow in the suction line (the larger of the circuit flow and the geometric pump delivery V_g·n)
[53] $Q_{prs} = Q_{line} * A_b / A_a$
= 61.988 L/min × 122.72 cm² / 59.101 cm²
= 128.71 L/min
Flow in the line to the cap port, worst stroke (retract: the cap-side discharge Q·A_b/A_a)
[54] $Q_{rtn} = Q_{line} * A_b / A_a$
= 61.988 L/min × 122.72 cm² / 59.101 cm²
= 128.71 L/min
Flow in the return line on the retract stroke (cap-side discharge)
[55] $v_{suc} = Q_{suc} / (pi * d_{suc}^2 / 4)$
= 65.25 L/min / (π × (50 mm)^2 / 4)
= 0.55386 m/s
Velocity in the suction line
[56] $u_{suc} = v_{suc} / v_{suc,max}$
= 0.55386 m/s / 1.2 m/s
= 0.46155
Velocity in the suction line over its limit
[57] $d_{min,suc} = sqrt(4 * Q_{suc} / (pi * v_{suc,max}))$
= sqrt(4 × 65.25 L/min / (π × 1.2 m/s))
= 33.969 mm
Smallest suction line bore for the velocity limit
[58] $d_{rec,suc}$ = smallest listed size not below the minimum bore
= d_{min} = 33.969 mm, 7 sizes listed
= 38 mm
Recommended suction line bore
[59] $v_{prs} = Q_{prs} / (pi * d_{prs}^2 / 4)$
= 128.71 L/min / (π × (25 mm)^2 / 4)
= 4.3702 m/s
Velocity in the pressure line

[60] $u_{prs} = v_{prs} / v_{prs,max}$
$= 4.3702\ m/s / 5\ m/s$
$= 0.87403$
Velocity in the pressure line over its limit
[61] $d_{min,prs} = sqrt(4 * Q_{prs} / (pi * v_{prs,max}))$
$= sqrt(4 \times 128.71\ L/min / (\pi \times 5\ m/s))$
$= 23.372\ mm$
Smallest pressure line bore for the velocity limit
[62] $d_{rec,prs}$ = smallest listed size not below the minimum bore
$= d_{min} = 23.372\ mm$, 7 sizes listed
$= 25\ mm$
Recommended pressure line bore
[63] $v_{rtn} = Q_{rtn} / (pi * d_{ret}^2 / 4)$
$= 128.71\ L/min / (\pi \times (32\ mm)^2 / 4)$
$= 2.6673\ m/s$
Velocity in the return line
[64] $u_{rtn} = v_{rtn} / v_{ret,max}$
$= 2.6673\ m/s / 3\ m/s$
$= 0.88911$
Velocity in the return line over its limit
[65] $d_{min,rtn} = sqrt(4 * Q_{rtn} / (pi * v_{ret,max}))$
$= sqrt(4 \times 128.71\ L/min / (\pi \times 3\ m/s))$
$= 30.174\ mm$
Smallest return line bore for the velocity limit
[66] $d_{rec,rtn}$ = smallest listed size not below the minimum bore
$= d_{min} = 30.174\ mm$, 7 sizes listed
$= 32\ mm$
Recommended return line bore
[68] $L_b = L_{closed}$ + stroke
$= 950\ mm + 800\ mm$
$= 1750\ mm$
Buckling length: the pin-to-pin length at full stroke, closed length plus stroke
[69] $L_{eff} = K_m * L_b$
$= 1 \times 1750\ mm$
$= 1750\ mm$
Effective length K·L
[70] $I_r = pi * d_r^4 / 64$
$= \pi \times (90\ mm)^4 / 64$
$= 3220600\ mm^4$
Second moment of area of the rod
[71] $r_g = d_r / 4$
$= 90\ mm / 4$
$= 22.5\ mm$
Radius of gyration of the solid rod
[72] $lam = L_{eff} / r_g$
$= 1750\ mm / 22.5\ mm$
$= 77.778$
Slenderness ratio
[73] $lam_c = pi * sqrt(2 * E / f_y)$
$= \pi \times sqrt(2 \times 210\ GPa / 500\ MPa)$
$= 91.052$
Transition slenderness between Johnson and Euler

[74] $F_{cr} = A_r * f_y * (1 - f_y * lam^2 / (4 * pi^2 * E))$
$= 63.617\ cm^2 \times 500\ MPa \times (1 - 500\ MPa \times 77.778^2 / (4 \times \pi^2 \times 210\ GPa))$
$= 2020.4\ kN$
Critical load, Johnson parabola (stocky rod)
[75] $F_{allow} = F_{cr} / S_f$
$= 2020.4\ kN / 3.5$
$= 577.25\ kN$
Allowable push load: critical load over the safety factor
[76] $F_{stall} = p * A_b$
$= 180\ bar\ g \times 122.72\ cm^2$
$= 220.89\ kN$
Stall thrust: the supply pressure on the bore area (no flow, so no back pressure; the friction of the seals does not relieve the rod of a stalled cylinder)
[77] $F_{buck} = max(F_{push}, F_{stall})$
$= max(150\ kN, 220.89\ kN)$
$= 220.89\ kN$
Rod load for the buckling check: the larger of the push load entered and the stall thrust $p \cdot A_b$
[78] $u_{buck} = F_{buck} / F_{allow}$
$= 220.89\ kN / 577.25\ kN$
$= 0.38267$
Buckling utilisation
[79] $u_{push} = F_{push} / F_{ext}$
$= 150\ kN / 209.85\ kN$
$= 0.7148$
Push load over the extend force
[80] $u_{pull} = F_{pull} / F_{ret}$
$= 60\ kN / 101.06\ kN$
$= 0.59369$
Pull load over the retract force
[81] $p_{push} = (F_{push} / \eta_m + p_{back} * A_a) / A_b$
$= (150\ kN / 0.95 + 0\ bar\ g \times 59.101\ cm^2) / 122.72\ cm^2$
$= 128.66\ bar$
Pressure at the cylinder the push load needs: $(F_{push}/\eta_m + p_{back} \cdot A_a)/A_b$

5. Results

SYMBOL	VALUE	DESCRIPTION
Q	60 L/min	Flow into the cylinder
p	180 bar g	Supply pressure
P_{pump}	21.13 kW	Pump shaft power
u_{buck}	0.38	Rod buckling utilisation (screening)
F_{ext}	209.85 kN	Extend force
dV	9.817 L	Swept volume per stroke (the larger)
p_{push}	128.66 bar	Pressure the push load needs at the cylinder
T_{shaft}	139.17 N·m	Pump shaft torque
n_{shaft}	1450 rpm	Pump shaft speed
Q_{prs}	128.7 L/min	Flow in the line to the cap port (worst stroke)
ρ	870 kg/m³	Fluid density
d_{prs}	25 mm	Pressure line inside diameter

SYMBOL	VALUE	DESCRIPTION
A_b	122.72 cm²	Bore area
A_r	63.62 cm²	Rod area
A_a	59.10 cm²	Annulus area
φ	2.0764	Area ratio A_b/A_a
F_{ret}	101.06 kN	Retract force
v_{ext}	79.86 mm/s	Extend speed
v_{ret}	165.8 mm/s	Retract speed
t_{ext}	10.02 s	Extend time
t_{ret}	4.825 s	Retract time
t_{cycle}	14.84 s	Cycle time
dV_{ext}	9.817 L	Extend swept volume
dV_{ret}	4.728 L	Retract swept volume
Q_p	61.99 L/min	Pump delivery
Vg_{req}	43.56 cm³/rev	Pump displacement the circuit needs
Q_{line}	61.99 L/min	Flow in the pressure and return lines before the area ratio
Q_{suc}	65.25 L/min	Flow in the suction line
Q_{rtn}	128.7 L/min	Flow in the return line
v_{suc}	0.5539 m/s	Suction line velocity
v_{prs}	4.370 m/s	Pressure line velocity (cap-port line, worst stroke)
v_{rtn}	2.667 m/s	Return line velocity
$d_{min,suc}$	33.97 mm	Smallest suction line bore
$d_{min,prs}$	23.37 mm	Smallest pressure line bore
$d_{min,rtn}$	30.17 mm	Smallest return line bore
$d_{rec,suc}$	38 mm	Recommended suction line bore
$d_{rec,prs}$	25 mm	Recommended pressure line bore
$d_{rec,rtn}$	32 mm	Recommended return line bore
L_b	1750 mm	Buckling length (pin to pin at full stroke)
L_{eff}	1750 mm	Effective length
lam	77.778	Slenderness
lam_c	91.052	Transition slenderness
F_{cr}	2020.4 kN	Critical load (screening)
F_{allow}	577.25 kN	Allowable push load (screening)
F_{stall}	220.89 kN	Stall thrust $p \cdot A_b$

SYMBOL	VALUE	DESCRIPTION
F_{buck}	220.89 kN	Rod load judged for buckling: the larger of the push load and the stall thrust
$buckling_{note}$	"screening only; ISO/TS 13725 method not implemented; stocky rod, Johnson parabola"	Rod buckling check
u_{push}	0.71	Push load over the extend force
u_{pull}	0.59	Pull load over the retract force
u_{pump}	0.97	Circuit flow over the pump delivery (governing)
u_{suc}	0.46	Suction line velocity over its limit
u_{prs}	0.87	Pressure line velocity over its limit
u_{rtn}	0.89	Return line velocity over its limit

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Pump delivery covers the circuit flow	$u_{pump} = 0.97$	$u_{pump} \le 1$	0.97	
PASS	Return line velocity within its limit	$u_{rtn} = 0.89$	$u_{rtn} \le 1$	0.89	[3](record)
PASS	Pressure line velocity within its limit, cap-port line at the larger stroke flow	$u_{prs} = 0.87$	$u_{prs} \le 1$	0.87	[3](record)
PASS	Extend force covers the push load	$u_{push} = 0.71$	$u_{push} \le 1$	0.71	
PASS	Retract force covers the pull load	$u_{pull} = 0.59$	$u_{pull} \le 1$	0.59	
PASS	Suction line velocity within its limit	$u_{suc} = 0.46$	$u_{suc} \le 1$	0.46	[3](record)
PASS	Rod buckling at the larger of the push load and the stall thrust, screening only; ISO/TS 13725 method not implemented	$u_{buck} = 0.38$	$u_{buck} \le 1$	0.38	[4](relations)

7. Applicability

WARNING records: tabulated values not taken from a reference record are E, f_y (Rod material); K_m (Mounting effective-length factor); S_f (Rod buckling safety factor); $v_{suc,max}$, $v_{prs,max}$, $v_{ret,max}$ (Line velocity limits); hoses (Hose and pipe size list). Enter each as one so its source, edition and licence travel with it. [record]

Defaults used (inputs left empty):
η_{hm}: not entered, so the default of 0.9 is used.
C_d: not entered, so the default of 0.62 is used.

8. Verification

```
Template fp.cylinder@1.4.0 Content hash e9d829e843189b74 Engine calc-engine/1.0.0
Known-answer cases passed 7/7 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

9. References

1. ISO/TS 13725.
2. Esposito, Fluid Power with Applications.
3. cylinder.line-limits.
4. Euler / Johnson column.